

PAIWA — Prior Authorization Intake Workflow Agent

Problem Brief · Forward Deployed Engineering Capstone · Synthetic Demo

TARGET USER

John, PA Intake Analyst at a health plan handling hundreds of requests daily. PAIWA pre-assesses each packet using a **deterministic policy engine + LLM evidence extractor**, providing status, route, confidence, and a plain-English explanation. John reviews the evidence and can **approve, reject, edit, escalate, or request provider follow-up**.
The system recommends; John decides.

PROBLEM

PA intake screening is **manual, repetitive, rules-based, and high-consequence**. Analysts must check required fields, document recency, treatment-duration thresholds, missing information, and routing.

Manual screening:

- Takes minutes per packet
- Varies between reviewers
- Scales linearly with volume
- Sends incomplete packets downstream when items are missed
- Can delay care when screening is overly strict

Intake screening is the bottleneck before downstream review can begin.

VALUE HYPOTHESIS

If PAIWA deterministically checks policy requirements, extracts clinical evidence, identifies missing/conflicting information, drafts provider questions, and routes cases — **while keeping every final decision with a human** — analysts can work from pre-assessed queues instead of raw packets.

Expected value: lower handling time + fewer missed-information errors, without surrendering decision authority.

SUCCESS METRICS

- **Primary — Intake handling time: Median time from queue entry to analyst decision; target is meaningfully below manual baseline. *Illustrative: ~35 operator-hours/day saved.***
- **Missing-info recall: ≥95% of genuinely missing items flagged.**
- **Routing accuracy: ≥95% of cases routed correctly.**
- **Escalation recall: 100% of malformed, adversarial, or conflicting cases escalated to a human.**
- **Override rate: Low; every analyst override is captured as an evaluation case.**

Results against the 30-case labeled golden set are reported in the AI Evidence artifact.

CONSTRAINTS

- **Synthetic data only** — no PHI
- **Policy YAML is source of truth**; LLM never determines policy or routing
- **Human approval required** for every outcome
- **Intake completeness only** — no medical-necessity or coverage decisions
- **Judge-runnable stack**: Python 3.11 + LangGraph + Streamlit; 40 synthetic packets
- OpenAI-compatible API optional; **deterministic fallback** when unavailable

ASSUMPTIONS

- Intake completeness is a distinct, existing PA workflow step
- Policies are machine-readable
- Humans review all finalized outcomes
- Provider responses re-enter intake for reassessment
- The **30-case eval set** represents key intake scenarios: complete, missing, stale, threshold, conflicting, malformed, duplicate, ambiguous, adversarial, and long-narrative
- **Primary persona**: Analyst; **ADMIN**: governance, evaluation, and audit

DESIGN PRINCIPLES

Primary metric = speed. Guardrails = safety.

The quantified value claim is explicitly illustrative, while every constraint and assumption maps to the demo implementation: **5 policies · 40 packets · 30 eval cases · fail-closed LLM · human approval.**